

Portfolio Construction Report

Introduction

This report presents three portfolios representing different investment philosophies. Each portfolio is constructed using eligible asset classes defined in the Investment Policy Statement and is intended to evaluate how asset allocation influences portfolio risk, return, and resilience under varying market conditions.

The strategic allocations presented below represent the initial target weights established at the outset of the project. As outlined in the Investment Policy Statement, these allocations are reviewed periodically and may be refined over time through the ongoing risk-monitoring and rebalancing framework, without changing a portfolio's underlying investment objective.

Portfolio A – Aggressive Growth

Objective

To maximize long-term capital appreciation by maintaining a growth-oriented asset allocation while accepting higher levels of volatility and larger drawdowns in pursuit of higher expected returns.

Portfolio Strategy

The portfolio emphasizes a high allocation to equity asset classes while maintaining diversification across sectors and geographic regions to reduce unnecessary concentration risk. Fixed-income exposure is intentionally limited to maximize long-term growth potential while accepting greater short-term volatility and market risk.

Risk Tolerance

This portfolio has a high risk tolerance and is designed for investors with a long-term investment horizon who are willing to accept significant short-term volatility and larger drawdowns in pursuit of higher expected long-term returns. While substantial market fluctuations are expected, diversification and ongoing risk monitoring are intended to reduce unnecessary concentration risk and maintain alignment with the portfolio's investment objective. Aggressive doesn't mean careless.

Strategic Asset Allocation

Asset Class	Allocation	Primary Purpose
U.S. Equity	55%	Long-term capital appreciation
International Equity	20%	Geographic diversification
Emerging Markets	10%	Higher long-term growth potential
Bonds	5%	Volatility reduction
Gold	5%	Inflation / Crisis hedge
Cash	5%	Liquidity

Portfolio Construction Rationale

The Aggressive Growth portfolio allocates the majority of capital to equities because they are expected to provide the greatest long-term capital appreciation. International and emerging market allocations broaden geographic exposure and reduce reliance on a single market. Modest allocations to bonds, gold, and cash provide stability, diversification, and liquidity without materially reducing the portfolio's long-term growth objective.

Implementation

Asset Class	Implementation Vehicle	Ticker	Selection Rationale
U.S. Equity	Vanguard Total Stock Market ETF	VTI	Broad exposure to U.S. equity market
International Equity	Vanguard Total International Stock ETF	VXUS	Broad international equity exposure
Emerging Markets	Vanguard FTSE Emerging Markets ETF	VWO	Broad emerging market equity exposure
Bonds	Vanguard Total Bond Market ETF	BND	Broad U.S. investment grade bond exposure
Gold	SPDR Gold Shares	GLD	Diversified gold exposure
Cash	iShares 0-3 Month Treasury Bond ETF	SGOV	Short term Treasury exposure and liquidity

Implementation Rationale

Implementation vehicles were selected to provide broad, low-cost, and highly liquid exposure to each asset class while minimizing implementation complexity. The same implementation vehicles are used across all three portfolios so that differences in portfolio performance primarily reflect strategic asset allocation decisions rather than differences in implementation or factor selection. Historical back testing analysis uses VEU and BIL as proxies for VXUS and SGOV, respectively, due to their longer trading histories; see the accompanying analysis notebooks for details.

Expected Strengths

Highest long-term return potential, Strong participation during sustained bull markets, Reduced geographic concentration risk, High expected capital appreciation

Expected Risks

Highest portfolio volatility, Larger potential drawdowns during market stress, greater sensitivity to equity market declines, higher VaR and Expected Shortfall than the other portfolios

Portfolio B – Balanced Growth

Objective

Achieve long-term capital appreciation while balancing growth opportunities with downside risk through diversified asset allocation. The portfolio is designed to provide attractive risk-adjusted returns while maintaining moderate levels of volatility and drawdown.

Portfolio Strategy

The portfolio maintains diversified exposure to equity asset classes while increasing allocations to defensive assets, including investment-grade bonds and cash equivalents, to reduce overall portfolio volatility. The strategy seeks to balance long-term growth with risk management by improving resilience during adverse market conditions without significantly limiting long-term return potential.

Risk Tolerance

This portfolio has a moderate risk tolerance and is designed to balance long-term capital appreciation with downside risk management. Moderate levels of short-term volatility and drawdowns are acceptable in pursuit of attractive risk-adjusted returns, while diversification and increased allocations to defensive assets seek to improve portfolio resilience during adverse market conditions.

Strategic Asset Allocation (Table)

Asset Class	Allocation	Primary Purpose
U.S. Equity	40%	Long-term capital appreciation
International Equity	15%	Geographic diversification and global growth exposure
Emerging Markets	10%	Higher long-term growth potential
Bonds	25%	Reduce portfolio volatility and support portfolio stability
Gold	5%	Diversification and Inflation / Crisis hedge
Cash	5%	Liquidity and portfolio flexibility

Portfolio Construction Rationale

The Balanced Growth portfolio seeks to balance long-term capital appreciation with downside risk management through a diversified allocation across growth-oriented and defensive asset

classes. While equities remain the primary source of long-term returns, increased allocations to investment-grade bonds provide greater portfolio stability and reduce overall volatility. Gold and cash provide additional diversification and liquidity while preserving the portfolio's long-term growth objective.

Implementation

Implementation vehicles are identical to those presented in Portfolio A. Holding implementation constant allows the analysis to isolate the effects of strategic asset allocation on portfolio risk and performance.

Implementation Rationale

The implementation rationale is identical to that presented in Portfolio A.

Expected Strengths

Improved risk-adjusted return potential, Lower portfolio volatility than the Aggressive Growth portfolio, Greater resilience during periods of market stress, Maintains meaningful long-term growth potential through diversified equity exposure

Expected Risks

Lower expected long-term returns than the Aggressive Growth portfolio, May underperform during strong bull markets due to increased defensive allocations, Still subject to market risk and economic downturns, Moderate VaR and Expected Shortfall relative to other portfolios

Portfolio C – Resilient Growth

Objective

Achieve long-term capital appreciation while emphasizing portfolio resilience, capital preservation, and downside risk management. The portfolio is designed to provide stable risk-adjusted returns through lower volatility and improved performance during adverse market conditions.

Portfolio Strategy

The portfolio maintains diversified exposure to equity asset classes while increasing allocations to high-quality defensive assets, including investment-grade bonds, gold, and cash equivalents, to improve resilience during periods of market stress. The strategy seeks to reduce portfolio volatility and downside risk while maintaining meaningful participation in long-term market appreciation through disciplined diversification.

Risk Tolerance

This portfolio has a lower to moderate risk tolerance and prioritizes resilience during periods of market stress while maintaining long-term growth potential. Lower levels of volatility and smaller drawdowns are preferred in exchange for greater portfolio stability, diversification, and capital preservation during adverse market conditions.

Strategic Asset Allocation (Table)

Asset Class	Allocation	Primary Purpose
U.S. Equity	25%	Long-term capital appreciation
International Equity	10%	Geographic diversification
Emerging Markets	5%	Additional long-term growth potential
Bonds	45%	Reduce portfolio volatility and support portfolio stability
Gold	10%	Diversification and Inflation / Crisis hedge
Cash	5%	Liquidity and portfolio flexibility

Portfolio Construction Rationale

The Resilient Growth portfolio is designed to improve portfolio stability and downside protection while maintaining meaningful long-term capital appreciation. A reduced allocation to equity asset classes is complemented by increased exposure to high-quality bonds and modest allocations to gold and cash to reduce portfolio volatility and enhance resilience during periods of market stress. This allocation reflects a long-term investment approach that prioritizes capital preservation and disciplined risk management while remaining invested in global financial markets.

Implementation

Implementation vehicles are identical to those presented in Portfolio A. Holding implementation constant allows the analysis to isolate the effects of strategic asset allocation on portfolio risk and performance.

Implementation Rationale

The implementation rationale is identical to that presented in Portfolio A.

Expected Strengths

Improved resilience during periods of market stress, Lower portfolio volatility, Smaller expected drawdowns, Strong downside risk management while maintaining long-term growth potential

Expected Risks

Lower expected long-term capital appreciation than Aggressive and Balanced portfolios, May underperform during sustained bull markets, Greater opportunity cost from increased defensive asset allocations, Reduced participation in higher-growth equity markets

Portfolio Comparison Summary (Making Predictions)

Metric	Aggressive	Balanced	Resilient
Expected Return	High	Moderate	Lower
Expected Volatility	High	Moderate	Lower
Expected Drawdown	High	Moderate	Lower
Diversification	Moderate	Good	Excellent
Market Stress Performance	Weakest	Moderate	Strongest

These predictions were directionally confirmed by the full risk and performance analysis; see Notebook 3 for realized results.